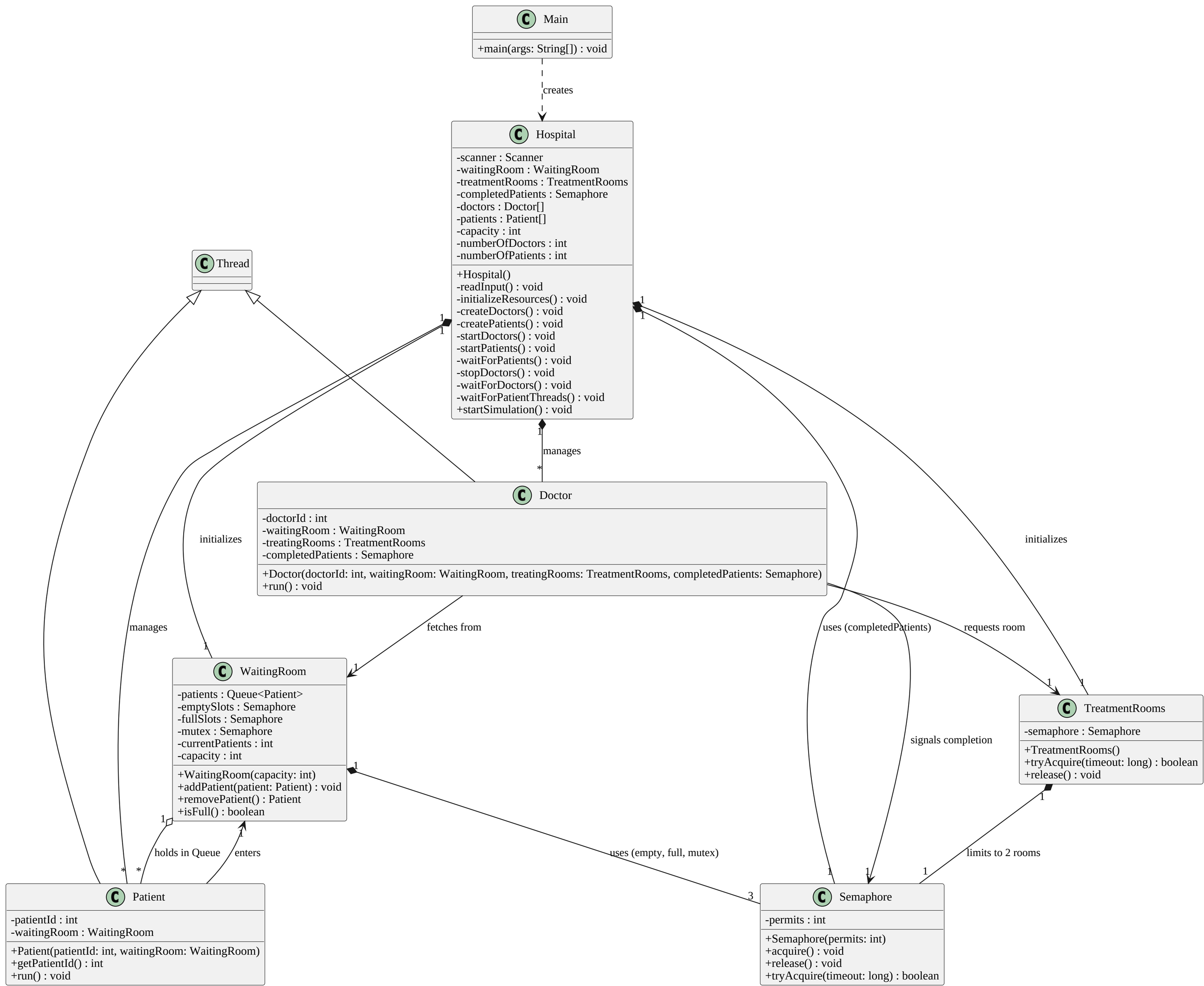
Main
+main(args: String[]) : void
creates
Hospital
-scanner : Scanner
-waitingRoom : WaitingRoom
-treatmentRooms : TreatmentRooms
-completedPatients : Semaphore
-doctors : Doctor[]
-patients : Patient[]
-capacity : int
-numberOfDoctors : int
-numberOfPatients : int
+Hospital()
-readInput() : void
-initializeResources() : void
-createDoctors() : void
-createPatients() : void
-startDoctors() : void
-startPatients() : void
-waitForPatients() : void
-stopDoctors() : void
-waitForDoctors() : void
-waitForPatientThreads() : void
+startSimulation() : void
Thread
manages
initializes
Doctor
-doctorId : int
-waitingRoom : WaitingRoom
-treatingRooms : TreatmentRooms
-completedPatients : Semaphore
+Doctor(doctorId: int, waitingRoom: WaitingRoom, treatingRooms: TreatmentRooms, completedPatients: Semaphore)
+run() : void
fetches from
uses (completedPatients)
requests room
WaitingRoom
-patients : Queue<Patient>
-emptySlots : Semaphore
-fullSlots : Semaphore
-mutex : Semaphore
-currentPatients : int
-capacity : int
+WaitingRoom(capacity: int)
+addPatient(patient: Patient) : void
+removePatient() : Patient
+isFull() : boolean
TreatmentRooms
-semaphore : Semaphore
+TreatmentRooms()
+tryAcquire(timeout: long) : boolean
+release() : void
signals completion
holds in Queue
enters
uses (empty, full, mutex)
limits to 2 rooms
Patient
-patientId : int
-waitingRoom : WaitingRoom
+Patient(patientId: int, waitingRoom: WaitingRoom)
+getPatientId() : int
+run() : void
Semaphore
-permits : int
+Semaphore(permits: int)
+acquire() : void
+release() : void
+tryAcquire(timeout: long) : boolean